Unit 6, Lesson 13

Situations Involving Equal-Size Groups

Standards

Building On	**3.OA.A.2, 3.OA.A.3**
Addressing	**4.NBT.B.6**
Building Toward	**4.NBT.B.6**

Instructional Routines

- Estimation Exploration
- MLR5 Co-Craft Questions

Goals

- Explain (orally and in writing) strategies for solving "how many in each group" and "how many groups" problems within 100.
- Interpret (orally) division equations that represent "how many in each group" and "how many groups" problems.

Student Facing Learning Goals

Let's solve division problems.

Lesson Purpose

The purpose of this lesson is for students to solve division problems in context, both those that ask "how many in each group" and those that ask "how many groups," in a way that makes sense to them.

Narrative

In IM Grade 3, students learned about the relationship between multiplication and division and reasoned about division in terms of equal groups. They also saw two interpretations of division: as a way to find the size of a group and as a way to find the number of groups. They used these understandings to reason about division problems in context, and to find a whole-number **quotient** from a two-digit **dividend** and a one-digit **divisor**.

In this lesson, students make sense of a division context and relate it to expressions. Students have the opportunity to represent and solve the problems in a way that makes sense to them, which allows teachers to observe the knowledge and understanding students have.

Access for Students with Disabilities

- Action and Expression

Access for English Learners

- MLR2

Lesson Timeline

Warm-up	10 min
Activity 1	15 min
Activity 2	20 min
Synthesis Estimate	10 min

Teacher Reflection Questions

What surprised you about the reasoning and strategies students used during this lesson? What assumptions do you have about student ability that are evidenced by what surprised you?

Warm-up

10 min

Estimation Exploration: Lots of Paletas

Standards

Building On **3.OA.A.3**

Instructional Routines

- Estimation Exploration

This *Estimation Exploration* prompts students to practice making a reasonable estimate based on experience and known information. In this case, it is not practical to count the paletas, but students could reason about groups of paletas by color, or estimate the complete rows and columns of paletas and extend their estimate to the whole set. Some students might also make an estimate based on their familiarity with how paletas are usually arranged in cases.

Student Task Statement

How many paletas (pah-LAY-tuhs) are in the case?



Record an estimate that is:

too low	about right	too high

Student Response

- Too low: less than 30
- About right: 30–80
- Too high: more than 80

Launch

- Groups of 2
- Display the image.
- "These are ice pops called paletas (pah-LAY-tuhs). They originated in Mexico and are typically made with many different fruits."
- Ask students to estimate without counting.
- "What is an estimate that's too high? Too low? About right?"
- 1 minute: quiet think time

Activity

- "Discuss your thinking with your partner."
- 1 minute: partner discussion
- Record responses.

Activity Synthesis

- "Is anyone's estimate less than 30? Greater than 80?"
- "How did you know that 30 (or another number) would be too low and 80 (or another number) too high?"
- "Based on this discussion does anyone want to revise their estimate?"

Activity 1

15 min

Paletas for a Class Party

Standards

Building On	**3.OA.A.2**
Building Toward	**4.NBT.B.6**

Instructional Routines

- MLR5 Co-Craft Questions

In this activity, students recall what they know about division from IM Grade 3. The context allows students to connect lived experiences to the math of the activity. By inviting students to consider treats that they enjoy in their homes or neighborhoods, they share experiences and foster connections that build community.

The first question gives students an opportunity to co-craft mathematical questions based on a situation before answering a question based on a division equation. Students divide a two-digit number by a one-digit divisor, as they did in IM Grade 3, in a way that makes sense to them. The *Activity Synthesis* highlights different representations students made and relates them to the situation. The terms **dividend** and **divisor** are re-introduced in this lesson to describe a number being divided into equal groups and a number by which another number (the dividend) is divided, respectively. Another term familiar to students is **quotient**, the result of a division expression.

This activity uses *MLR5 Co-craft Questions*. Advances: writing, reading, representing

Access for Students with Disabilities

Action and Expression: Develop Expression and Communication. Give students access to concrete manipulatives (connecting cubes, counters, or square tiles) and grid paper. Invite students to use these to act out or draw $84 \div 7$. *Supports accessibility for: Conceptual Processing, Attention*

Student Task Statement



Diego's aunt buys paletas, which are ice pops, for a class party. At the local market, paletas come in different flavors. She buys the same number of paletas of each flavor.

1. What mathematical questions can we ask about this situation?
2. Here is an equation: $84 \div 7 = ?$

   In the situation about the class party, what questions could the equation represent?
3. Find the answer to one of the questions you wrote. Explain or show your reasoning.

Launch

- Groups of 2
- "What are some of your favorite treats or snacks from home?"
- 30 seconds: quiet think time
- 1 minute: partner discussion

MLR 5: Co-craft Questions

- Display the opening paragraph and the first question.
- "Write a list of mathematical questions that could be asked about this situation."
- 2 minutes: independent work time
- 2–3 minutes: partner discussion
- Invite several students to share one question with the class. Record responses.
- "How are these questions alike?" (The questions involve multiplying or dividing.) "How are they different?" (The questions can be answered using

Student Response

1. Sample responses:
   - How many people are coming to the party?
   - How many flavors of paletas are there?
   - How many paletas of each flavor did Diego's aunt buy?
   - How many flavors of paletas did she buy?
   - How much money does she have to spend on paletas?
2. Sample responses:
   - Diego's aunt is buying 84 paletas in 7 flavors. How many paletas of each flavor is she buying?
   - Diego's aunt is buying 84 paletas. There are 7 paletas of each flavor. How many flavors are there?
3. $84 \div 7 = 12$. Sample responses:
   - I know that 7 groups of 10 is 70 and 7 groups of 2 is 14, so 7 groups of 12 is 84.
   - $10 \times 7 = 70$ and $2 \times 7 = 14$, so $12 \times 7 = 84$.
   - $70 \div 7 = 10$ and $14 \div 7 = 2$, so $84 \div 7 = 12$.

different operations.)

- "Let's look at the next question in the activity."

Activity

- 3–4 minutes: quiet work time

Activity Synthesis

- Display two questions that students wrote for the equation $84 \div 7 = ?$.
- "What does the 84 represent in both problems?" (The amount being divided into equal groups.)
- "In mathematics, the number being divided is known as the **dividend**, and the number by which the dividend is divided is known as the **divisor**. The result of the division is known as the **quotient**."
- Invite students to share their strategies for the last question. Highlight strategies that show equal-size groups and reasoning that relates multiplication and division.

Advancing Student Thinking

If students are unsure how to represent $84 \div 7 = ?$ with a question about the class party situation, consider asking:

- "If you were to act out the meaning of this equation, what would the 84 represent? What would the 7 represent? What would the ? represent?"
- "How can you use what you shared to write a question that matches the equation?"

Activity 2

20 min

More Snacks for a Class Party

Standards

Building On	**3.OA.A.3**
Addressing	**4.NBT.B.6**

The purpose of this activity is for students to continue to use any representation or strategy to solve division problems with two- and three-digit dividends. They solve one problem that asks how many in each equal group and another that asks how many equal groups.

Student Task Statement

1. Priya's mom makes 85 gulab jamuns for the class to share. Priya gives 5 gulab jamuns to each student in the class.



   How many students are in Priya's class? Explain or show your reasoning.

2. Han's uncle sends 108 chocolate-covered breadsticks for a snack. The students in Han's class are seated at 6 tables. Han plans to give the same number of breadsticks to each table.



   How many breadsticks does each table get? Explain or show your reasoning.

Launch

- Groups of 2
- Display images of snacks in the activity.
- "Take a look at the images. What do you notice? What do you wonder?"
- Share responses.
- "Gulab jamuns are sweet treats that are popular in India, Pakistan, and their neighboring countries in South Asia."
- "Breadsticks that are covered with chocolate, strawberry cream, or other flavors are popular snacks in Japan, Taiwan, and other East Asian countries."
- Ask a student to read the first problem aloud.
- "The problems in this activity involve treats that students enjoy from different places around the world."

Student Response

1. 17 students. Sample response: I know that 50 is 10 groups of 5, and then there are 35 more gulab jamuns to give out. I know that 35 is 7 groups of 5. So, Priya gave out gulab jamuns to $10 + 7$ kids.
2. 18 breadsticks. Sample responses: $108 = 60 + 48$.
   - Sixty is 6×10, and 48 is 6×8. Adding 10 and 8 gives 18.
   - $60 \div 6 = 10$ and $48 \div 6 = 8$. So, $10 + 8 = 18$.

Activity

- 6–8 minutes: independent work time
- 2–3 minutes: partner discussion

Activity Synthesis

- "How are the questions in the two situations—gulab jamuns and breadsticks—alike?" (They involved division into equal groups.)
- "How are they different?" (The first looks for the number of groups. The second looks for the size of a group.)
- Invite students to share their response for the second problem.
- "What is the same about each of these methods? What is different?"

Lesson Synthesis

"Today we solved problems involving division of whole numbers. We thought about the kinds of division problems we were solving—whether we were trying to find the number of groups or the amount in each group."

Display: One-hundred four students are put into groups $104 \div 8 = ?$

"In this situation, what questions could the equation represent?" (How many students are in each group if 104 students

are put into 8 equal groups? How many groups can be made if 104 students are put into groups of 8?)

"What does the value of $104 \div 8$ mean?" (It depends on the question being answered: either the number of students or the number of groups.)

Suggested Centers

- Five in a Row: Multiplication (3–5), Stage 3: Two-Digit Factors (Addressing)
- Compare (1–5), Stage 4: Divide within 100 with One-Digit Divisors (Supporting)

Cool-down

5 min

Cleaning Desks

Standards

Addressing **4.NBT.B.6**

Student Task Statement

A group of students are cleaning 96 desks after an art project. If 6 students each clean the same number of desks, how many desks does each student clean? Explain or show your reasoning.

Student Response

Each student cleans 16 desks. Sample responses:

- $6 \times 10 = 60$ and $6 \times 6 = 36$ so $6 \times 16 = 96$.
- $60 \div 6 = 10$, $36 \div 6 = 6$, and $10 + 6 = 16$.

Responding to Student Thinking

Students show they understand the situation as a division problem, but find a quotient other than 16.

Next Day Supports
After the *Warm-up* in the next lesson, pair students up to discuss their responses and give students access to base-ten blocks.

The work in this lesson builds from the division concepts developed in a prior unit.

Prior-Unit Support
Grade 3, Unit 4, Section A What Is Division?